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Bayesian Optimization of Multi-Material 3D-Printed Tensegrity-Inspired Structures for Energy Absorption

Tensegrity-inspired architectures combine rigid compression members with a continuous network of soft tension elements, yielding lightweight assemblies with tunable nonlinear responses that are attractive for energy-absorbing applications such as protective equipment, packaging, and aerospace landing structures. Multi-material fused-deposition modeling (FDM) of polylactic acid (PLA) struts and thermoplastic polyurethane (TPU) tension elements makes it possible to fabricate diverse tensegrity-inspired geometries rapidly enough to support a high-throughput experimental search, while the rate-dependent damping of TPU is itself an asset for impact mitigation. We report an experiment-driven Bayesian optimization (BO) framework in which a Gaussian-process surrogate is trained directly on physical quasi-static compression and drop-weight impact data and used to recommend the next batch of designs to print and test. The framework parameterizes a family of tensegrity-inspired unit cells over strut geometry, tension-element cross-section, connectivity topology, and unit-cell tiling, and optimizes peak transmitted force, specific energy absorption, and compaction efficiency.

Keywords: tensegrity, multi-material 3D printing, Bayesian optimization, energy absorption, design of experiments

1 Introduction

Tensegrity structures are assemblies of rigid compression members suspended within a continuous network of tension members; their defining property, that no two rigid bars touch, yields lightweight assemblies with remarkable strength-to-weight ratios and tunable nonlinear mechanical responses [1,2]. These properties have made tensegrity attractive across scales, from metamaterial unit cells [3–5] to compliant impact-absorbing robots developed for planetary landing [6–10]. Recent work has shown that *tensegrity-inspired* unit cells can be directly 3D-printed and exhibit load-limiting plateaus under drop-weight impact [11], opening a path toward additively manufactured architected absorbers. Throughout, we use *tensegrity-inspired* to denote printable unit cells that adopt the characteristic tensegrity motif—rigid compression members carried by a network of pretensioned soft tension elements—while relaxing the strict classical requirement that no two compression members touch and that the tension members behave as ideal inextensible cables; here the tension elements are extruded TPU and the struts may share printed junctions.

Multi-material fused-deposition modeling (FDM) that pairs a rigid polymer such as polylactic acid (PLA) with a soft, rate-dependent elastomer such as thermoplastic polyurethane (TPU) has been used to fabricate rigid–soft architected energy absorbers on a single platform—for example, thick-panel origami metamaterials with PLA panels and TPU hinges [12] and ABS/TPU honeycombs with tunable stiff/flexible proportions [13]. The same rigid–soft co-printing capability makes it possible to rapidly fabricate diverse tensegrity-inspired geometries, which adopt a discrete-compression/continuous-tension topology rather than an origami or honeycomb one. Although extruded TPU does not behave as an idealized inextensible cable, its rate-dependent damping is a desirable property for impact-absorption applications and complements the elastic stiffness of the PLA compression members rather than competing with it.

Designing such architected absorbers is fundamentally a *design-of-experiments* problem: the configuration space spanned by strut geometry, tension-element cross-section, connectivity, and unit-cell tiling is large; each candidate must be physically printed and tested; and the relevant performance metrics—peak transmitted force, specific energy absorption (SEA), and compaction efficiency—are noisy and partially correlated. Bayesian optimization (BO) is well-matched to this regime, having driven breakthroughs from hyperparameter tuning of AlphaGo [14] to sustainable concrete [15], autonomous discovery of organic laser emitters [16], and architected materials [17–20].

1.0.1 Contributions.. This paper makes the following contributions:

(1) A parameterized family of multi-material 3D-printable tensegrity-inspired unit cells with PLA compression members and TPU tension elements that are anchored *inside* the ends of each PLA strut—the strut acting as a rigid cage in which the cables meeting at a given end join before exiting through discrete ~~outlets—to ensure~~ outlets—a joint geometry intended to improve cyclic interface durability, to be verified by pull-out and fatigue testing.
(2) An experiment-driven BO loop, built on BoTorch/Ax [21,22], that operates directly on physical quasi-static compression and drop-weight impact measurements, recommending the next designs to fabricate.
(3) High-fidelity validation experiments using *pretensioned* tensegrity assemblies (true cables and measured pretension) for the Pareto-optimal designs identified by the single-stage loop above, reported separately rather than fused into the optimization.

The remainder of the paper is organized as follows. Section 2 reviews tensegrity mechanics, multi-material 3D printing for energy absorption, and Bayesian optimization with an emphasis on

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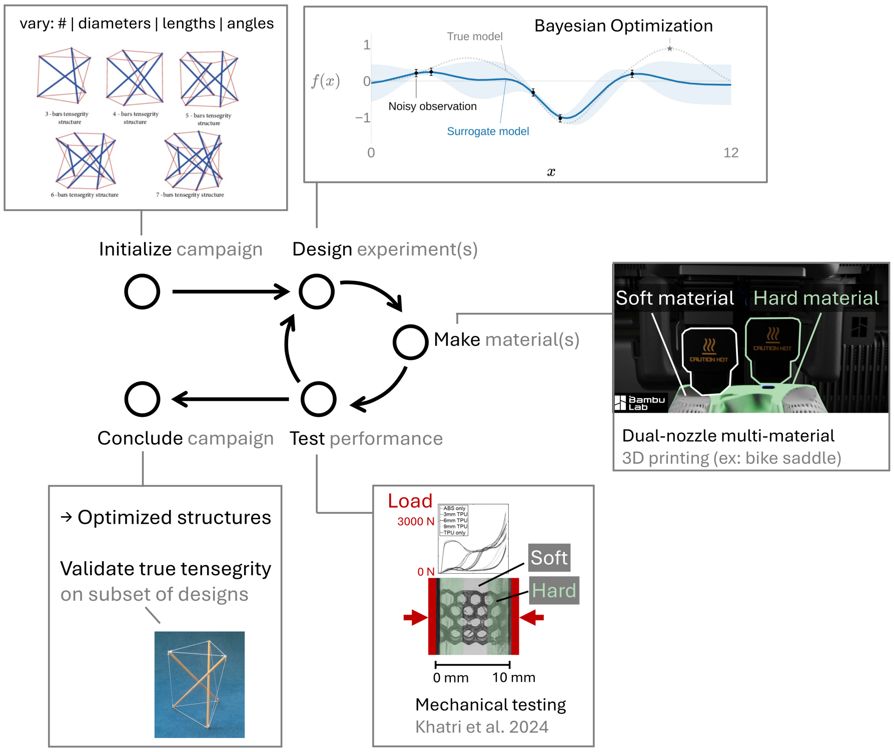


Fig. 1 Closed-loop, experiment-driven design framework. Parameterized PLA/TPU tensegrity-inspired unit cells are printed, tested under quasi-static compression and drop-weight impact, and the resulting performance data drive a Gaussian-process surrogate that recommends the next batch of designs to fabricate.

multi-objective and noise-robust acquisition functions. Section 3 introduces the design parameterization, fabrication workflow, experimental protocols, and BO loop. Section 4 reports the results of the closed-loop experimental campaign, and Section 5 discusses the implications and limitations of the approach. Section 6 concludes.

2 Background

2.1 Tensegrity-Inspired Architectures for Energy Absorption. Tensegrity prisms and their planar extensions exhibit softening–stiffening response under compression [3–5], characteristics desirable for impact mitigation because the structure can absorb a large fraction of the input energy at a controlled, sub-injurious force plateau. Santos [23] recently characterized a tensegrity energy-dissipation metamaterial reporting equivalent viscous damping on the order of ~23% with negligible residual strain per impact, underscoring the suitability of tensegrity-inspired architectures for repeated-impact use cases. The space program literature documents large-scale tensegrity-inspired absorbers for planetary landing [6–10] and a long history of crushable energy-absorbing landing systems [24–26]. Pajunen et al. [11] demonstrated that truncated-tetrahedral tensegrity-inspired unit cells can be directly 3D-printed and exhibit load-limiting plateaus, motivating the present focus on additively manufactured TPU/PLA composites. More recent work on dynamically loaded additively-manufactured energy-absorbing lattices [27], together with Intrigila et al.'s bistable tensegrity-like unit cell for lattice metamaterials [28] as the closest published analog, confirms that this architecture class also responds favorably under impact and quasi-static loading, although neither study employed multi-material ~~FFF~~ FDM or closed-loop design optimization. Table 1 situates the present work against the three most closely related prior efforts, making explicit that the combination of a multi-material FDM tensegrity-inspired architecture with an *experiment*-driven optimization loop is, to our knowledge, not addressed by any single prior study.

2.2 Multi-Material 3D Printing of PLA/TPU Composites. Multi-material rigid–soft FDM enables stiff/compliant combinations on a single platform with tunable energy absorption, demonstrated for thick-panel origami with PLA (or ABS/CFRP) panels and TPU hinges [12] and for ABS–TPU honeycombs [13]. Neither study is itself a tensegrity, but both establish the rigid–soft co-printing route we adopt. Ye et al. [12] introduced a *wrapping-based* strategy in which rigid cores are encapsulated by continuous soft skins, ~~preventing interface delamination and enabling cyclic durability~~reported to reduce interface delamination under cyclic loading, something we take inspiration from in our designs. Recent multi-material PLA/TPU sandwich and layered composites [29,30] report quantitative bounds on stiffness, strength, and energy absorption for the same PLA–TPU pair used here, while ~~fused filament fabrication (FFF)~~ FDM PLA fatigue data [31,32] inform conservative design bounds for the rigid struts. Architected multi-material lattices with co-printed compliant skins have been explicitly designed for high specific energy absorption [33], with ~~FFF-specific~~ FDM-specific multimaterial interface adhesion (notably TPU bonded to rigid co-printed substrates) characterized in [34]; recent surveys catalogue the rapidly expanding additively manufactured polymeric energy-absorption literature [35,36].

2.3 Bayesian Optimization for Architected-Material Design. Bayesian optimization is a sequential, model-based strategy for optimizing expensive black-box functions using a Gaussian-process surrogate [37,38]. Modern implementations such as BoTorch [21] and the Ax platform built atop it [22] support parallel batched evaluations and a wide variety of acquisition functions; recent algorithmic advances include log-EI for numerical robustness [39], noisy expected hypervolume improvement for multi-

Table 1 Positioning of this work against the most closely related prior efforts. The present study is distinguished by combining a multi-material (PLA/TPU) FDM tensegrity-inspired architecture with a closed-loop optimizer trained directly on physical measurements rather than simulations or a single fabrication condition.

Study	Architecture	Fabrication	Optimization	Ground truth
Pajunen et al. 2019 [11]	Truncated-tetrahedral tensegrity-inspired	Single-material 3D print	None (single condition)	Experiment (drop impact)
Intrigila et al. 2022 [28]	Bistable tensegrity-like lattice unit cell	Single-material 3D print	None	Experiment (quasi-static/impact)
Mo et al. 2023 [17]	Architected lattices	— (simulation)	Multifidelity BO	Simulation (+ sparse high-fid.)
This work	Tensegrity-inspired T_3 prism (PLA/TPU)	Multi-material FDM (PLA + TPU)	Experiment-driven BO (SAASBO/qNEHVI)	Physical experiment

objective settings [40], input-noise-robust acquisitions [41], and evolution-guided constrained multi-objective BO for self-driving labs [42]. Mixed/categorical search ~~spaces—needed here for the discrete~~ spaces—needed for the planned connectivity-topology ~~variable—have~~ extension rather than the continuous T_3-prism prototype studied here—have dedicated treatments in BOCS [43] and the one-hot-with-rounded-kernel construction of [44]. BO has been applied to mechanical metamaterial and structural design [17–19] and to additive manufacturing [20]. Mo et al. [17] in particular established a multifidelity BO framework for architected materials by fusing low-fidelity simulations with high-fidelity ground truth via nonlinear information-fusion priors [45]; the present work shifts the emphasis from simulation surrogates to direct experimental data, appropriate for materials whose constitutive response is difficult to calibrate from first principles.

3 Materials and Methods

3.1 Design Parameterization. We parameterize a family of tensegrity-inspired unit cells via four groups of design variables. For the working T_3-prism prototype that defines the initial BO search space (Section 3.4), only the first two groups are active and all of their variables are *continuous*; the connectivity-topology and tiling groups are held fixed for this prototype and are listed here as planned extensions of the design family:

- **PLA compression members.** Strut diameter d_s and length ℓ_s, with the slenderness ratio ℓ_s/d_s bounded away from buckling-dominated regimes.
- **TPU tension elements.** Cross-sectional area A_t (or equivalent diameter d_t for circular cross-sections).
- **Connectivity topology (future extension).** Selected from a discrete set of candidate unit cells (e.g., truncated-tetrahedral, prism-based, and octahedral variants); when activated, the design vector ~~encodes~~ would encode this as a categorical choice. It is held fixed at the T_3 prism for the present campaign.
- **Unit-cell tiling (future extension).** Number of cells $N_x \times N_y \times N_z$ and the orientation of each cell relative to the impact direction; held fixed at a single cell here.

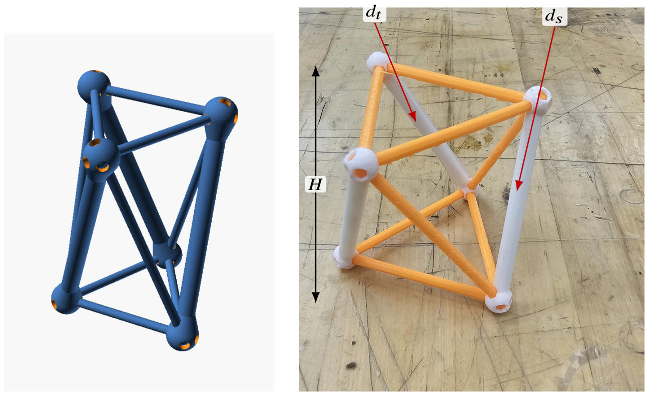
3.1.1 Working prototype. The first instance of this family used to validate the fabrication and test pipeline is a three-strut (T_3) tensegrity prism scaled to a ~50 mm bounding box~~, with d_s = and a cable~~. As implemented in the BO harness, this working prototype exposes five continuous design variables—base radius R, cell height H, twist angle θ, a single strut diameter d_s, and a single cable (tendon) diameter d_t~~swept over the FFF-resolvable set {1.2, 1.8, 2.4, 3.0, 4.5}~~—over the ranges listed in Table 2; the cable diameter d_t is treated as a *continuous* variable on $[3.0, 5.5]$ mm rather than a discrete set. Under the prism's D_3 symmetry ~~the twelve~~ all struts share one diameter and all cables share one diameter, so the present parameterization uses a single strut-diameter axis and a single cable-diameter axis (two member-diameter axes~~collapse to four orbit axes (one strut orbit and three cable orbits:~~), leaving room to relax to per-orbit (saddle, top, bottom) ~~, which keeps the initial design vector low-dimensional while leaving room to relax to~~ or fully per-member diameters once the ~~saddle/top/bottom~~ individual orbits have been ~~individually characterized.~~ characterized. Figure 2 shows the parametric CAD model alongside an as-printed prototype of this working T_3 prism.

Table 2 Design variables for the T_3-prism Sobol initialization batch, as implemented in the BO harness (single-iteration $n = 9$ Sobol set). The joint diameter is held fixed at the `t3-prism.scad` default; all specimens are printed in the vertical orientation and packed 3 × 3 on a single Bambu Lab H2D plate.

Design variable	Symbol	Units	Range
Base radius	R	mm	25–40
Cell height	H	mm	60–110
Twist angle	θ	deg	40–80
Strut (PLA) diameter	d_s	mm	6.0–12.0
Cable (TPU) diameter	d_t	mm	3.0–5.5
Joint diameter (fixed)	d_j	mm	7.0

3.2 Multi-Material Fabrication. Specimens are fabricated on a multi-material FDM printer using PLA for the rigid struts and TPU for the soft tension elements. Figure 3 summarizes the end-to-end fabrication and characterization workflow, from parametric CAD through multi-material slicing, dual-extrusion printing, post-processing, and mechanical testing. Rather than wrapping the struts in a continuous TPU skin, the current design anchors the TPU tension elements *inside* the ends of each PLA strut: the cables converging at a given strut end are joined within the strut, which acts as a rigid cage housing the junction, and the individual tendons then exit through discrete outlets. This keeps the soft–rigid interface in internal, compression-dominated pockets rather than at an exposed overmolded surface. This construction effectively inverts the material roles of Ye et al.'s soft-skin-over-rigid-core wrapping [12]: here a soft TPU core is captured within a rigid PLA shell, so we cite that work only as a multi-material co-printing precedent rather than as validation of the present junction.

3.2.1 Print platform and joint geometry.. The working platform is a Bambu Lab H2D printer, which permits a single rigid–soft build without filament swaps. Strut endpoints are tied to cables through parametric joint geometries developed and ranked through a five-design OpenSCAD study (anchor-bulb, dovetail, TPU-sleeve overmold, eyelet-loop, and TPU-rebar variants); the working prototype uses a dovetail joint (Design B) with an anchor-bulb backup (Design A) for sensitivity studies, with a captive TPU core routed inside the PLA shell to keep the soft tendon protected from layer-line failure at the strut-to-cable transition. The full five-design comparison and the selected junction are detailed in the Supplementary Information.



(*a*) (*b*)

Fig. 2 Working T_3-prism prototype. *(a)* Parametric OpenSCAD model whose base radius R, cell height H, twist angle θ, strut diameter d_s, and cable diameter d_t are the design variables of Table 2. *(b)* A representative dual-material print from the PR #35 print history (rigid PLA struts, here in white; tendon members in orange), with callouts marking the strut diameter d_s, a tendon (cable) diameter d_t, and the cell height H; the inter-plate twist θ is the relative rotation of the top and bottom node triangles, and the base radius R sets their circumradius.

3.2.2 Supports for soft members.. Because near-vertical TPU tendons are otherwise unsupported during printing, the slicer profile combines a tensegrity-specific Bambu Studio recipe (support threshold angle dropped from 40° to 10°, support material matched to the rigid extruder) with manually generated narrowing pillars that ray-cast against the printable mesh underside, producing TPU-safe coverage of vertical members without slicer-side painting. Full workflow details are provided in the Supplementary Information.

Table 3 Multi-material print parameters on the Bambu Lab H2D. Cells marked *TBD* are pending confirmation; the reporting structure follows the methods section of Ye et al. [12].

Parameter	PLA (rigid struts)	TPU (soft tendons)
Nozzle temperature (°C)	*TBD*	*TBD*
Bed temperature (°C)	*TBD*	
Layer height (mm)	*TBD*	
Print speed (mm/min)	*TBD*	*TBD*
Infill (%)	*TBD*	*TBD*
Build-plate type	Textured PEI	
Build orientation	Vertical	
Supports	Off in slicer; manual narrowing pillars	

3.3 Experimental Characterization.

3.3 Experimental Characterization. Each printed specimen is subjected to two physical tests:

3.3.1 Quasi-static compression.. A uniaxial load–displacement curve is recorded on a benchtop electromechanical load frame (BYU CB 123 Instron) following best-practice combinations of ASTM D638, ASTM D412, ASTM E111, ASTM F2971, and ISO/ASTM 52900/52921 adapted to architected polymer cells. Because expected cell stiffnesses fall in the ~5–30 g force range, the default 5 kN load cell is over-ranged and is being replaced by a 100–500 N cell for the first campaign. Reported metrics are

$$\mathrm{SEA} = \frac{1}{m}\int_0^{\delta_d} F(\delta)\,\mathrm{d}\delta, \qquad \eta_c = \frac{\int_0^{\delta_d} F(\delta)\,\mathrm{d}\delta}{F_{\max}\,\delta_d}, \tag{1}$$

where m is the specimen mass, δ_d is the densification displacement, $F_{\max}$ is the peak transmitted force, and η_c is the compaction efficiency. To keep these metrics objective and reproducible, the densification displacement δ_d is taken as the displacement at which the energy-absorption efficiency–displacement curve reaches its maximum ($\mathrm{d}\eta/\mathrm{d}\delta = 0$), and $F_{\max}$ is the maximum force recorded *within* the integration window $[0, \delta_d]$ rather than a global or post-densification peak.

3.3.2 Drop-weight impact.. Following Pajunen et al. [11], specimens are subjected to drop-weight impact tests with instrumented load cells; we report the peak transmitted force and the energy-absorption plateau characteristics. Figure 4 illustrates the mechanism-oriented presentation we adopt for these processed deceleration traces, linking features of the measured signal to the structural events that produce them. The primary fixture is a bungee-assisted laboratory drop tower in which the base accelerates downward faster than $1g$; because unconstrained specimens lift off the base during descent by design of the rig, the protocol constrains the specimen top through light tethers (capping upward travel) and registers the base via transfer tape or V-block features so that loading-direction compliance is not altered. A higher-fidelity replicate campaign on a Lansmont M23 shock/cushion tester (per ASTM D1596) with synchronized Polytec QTec laser vibrometry is planned for selected Pareto-optimal designs. Slip resistance and traction at the specimen–floor interface—governed by the tip geometry rather than the internal lattice—follow the test methodology of ISO 11334-4 [46] for walking-aid tips and are out of scope for the present axial-impact study.

3.3.3 Per-modality objectives.. Five complementary data streams (triaxial accelerometer, high-speed video, shaker transfer function, slug-firing gas gun, and Polytec QTec laser-Doppler vibrometer) each contribute a distinct raw observable; the mapping from each stream to BO-ready scalar objectives (SEA, VEA, $F_{\max}$, cushion-curve intercept, Gibson–Ashby relative density) is defined in five companion briefs and consolidated in a cross-modality synthesis.

3.4 Experiment-Driven Bayesian Optimization Loop. The optimization loop maintains a Gaussian-process (GP) surrogate over the design space defined in Section 3.1. After each round of physical experiments, the GP is updated and an acquisition function is maximized to recommend the next batch of designs to fabricate. Implementation uses BoTorch [21] with the log-Expected-Improvement (LogEI) acquisition [39] for the single-objective baseline and noisy Expected Hypervolume Improvement (qNEHVI) [40] for the multi-objective case (simultaneously maximizing SEA and compaction efficiency while constraining peak transmitted force). When experimental noise dominates (e.g., between-print variability), we additionally evaluate the input-noise-robust acquisition of Daulton et al. [41] and the evolution-guided constrained variant of Low et al. [42]. The ~~discrete connectivity-topology variable is handled with~~ working T_3-prism search space is fully continuous (the five variables of Table 2), so no categorical encoding is required for the present campaign; the mixed-search-space ~~construction~~ constructions of [44] ~~; and~~ and the BOCS combinatorial baseline [43] ~~is reported for comparison~~are noted as the route for the future connectivity-topology categorical (Section 3.1) rather than as part of the current loop.

The GP surrogate uses a Matérn-5/2 kernel with automatic relevance determination (ARD), i.e. a separate length scale per input dimension, so that per-variable sensitivities can be read directly from the fitted length scales. Inputs are normalized to the unit cube and outputs are standardized; observation noise is *inferred* by the model (a learned homoscedastic noise level) rather than fixed a priori. The campaign budget is $T = 10$ sequential batches of $q = 5$ prints each (50 specimens total), initialized with the $n = 9$ Sobol batch of Table 2. Random seeds for the Sobol

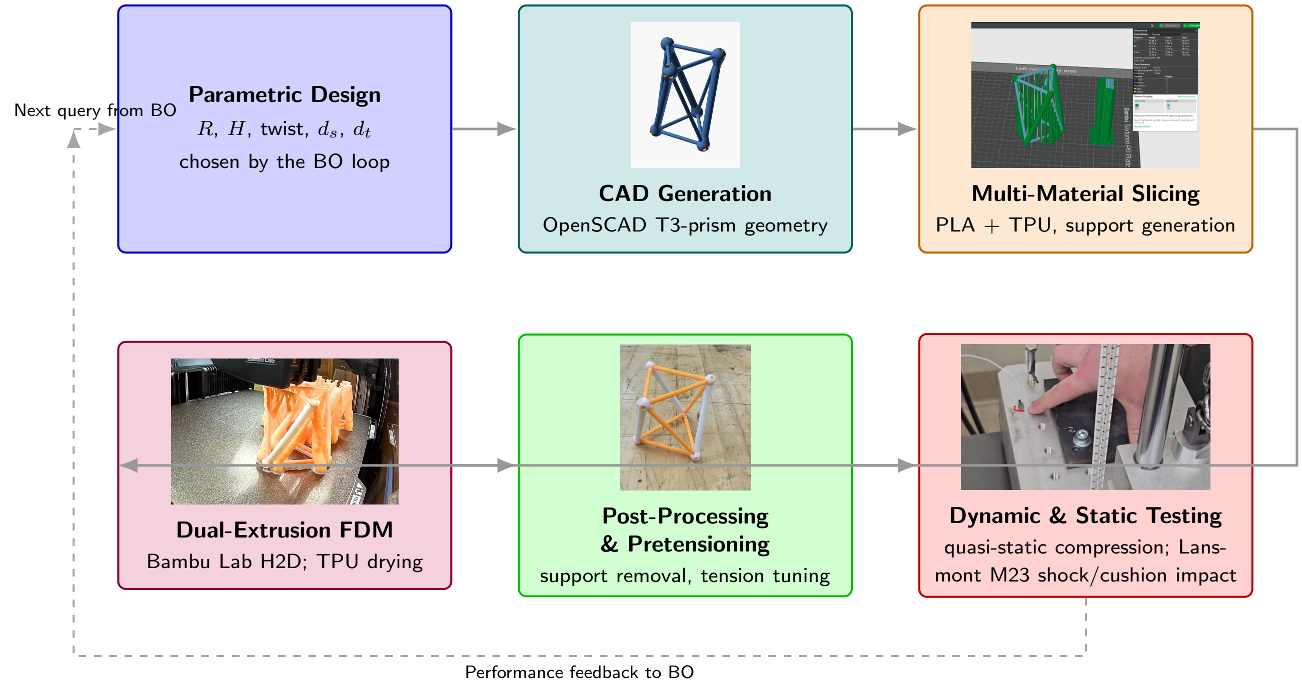


Fig. 3 Fabrication and characterization workflow for the multi-material tensegrity-inspired unit cells: design parameters drive parametric CAD, slicing with manually generated TPU supports, a single multi-material print on the Bambu Lab H2D, post-processing and inspection, and mechanical testing. Node photographs are representative artifacts drawn from the project's build history: the OpenSCAD T_3-prism render (CAD), the dual-nozzle Bambu Studio slice with PLA on the left extruder and TPU on the right (slicing), an in-progress H2D print of the prism batch (~~FFF~~FDM), an as-printed specimen (post-processing), and the bungee-assisted drop-tower base plate with accelerometer (testing).

generator and the acquisition optimizer are recorded per batch so that each recommendation is reproducible.

3.4.1 Implementation status.. The campaign harness is built on the Ax platform [22]. The working T_3-prism search space is the five-variable continuous box of Table 2; Ax's `HierarchicalSearchSpace` ~~so that printer-feasibility constraints (e.g. the cable-diameter categorical depending on the chosen connectivity topology) can be encoded directly in the search space rather than enforced post hoc~~machinery is reserved for the planned topology/tiling extensions (Section 3.1) and is not exercised by the current continuous-only prototype. The first physical batch consists of $n = 9$ specimens generated by a Sobol sequence over ~~the four D_3-orbit axes of the working T_3 prism~~ these five variables and printed in a single 3×3 plate layout. For ~~the present $\leq$25-dimensional parameterization~~ this ~5–6-variable space the default acquisition pairing is sparse axis-aligned subspace BO (SAASBO) with qNEHVI. We retain SAASBO even though the dimensionality is modest because its strong priors on length scales tend to improve uncertainty quantification and predictive accuracy at the small sample sizes typical of physical campaigns, not only in high-dimensional settings; escalation to trust-region BO (TuRBO) is planned only if larger per-member-diameter tilings push the design space well beyond this regime. The campaign code and recommended batches will be archived at a Zenodo-minted DOI mirroring the GitHub repository.

Once the campaign produces model data, the fitted surrogate is audited with the leave-one-out cross-validation and parameter-sensitivity diagnostics of Figs. 5 and 6.

3.5 Metal-Analog Validation. To test whether the rankings learned on the printed PLA/TPU specimens generalize beyond the as-printed polymer system, a final validation campaign reproduces a subset of the T_3-prism designs as *metal analogs* built from hollow aluminum rods (compression members) and stainless-steel threaded cables (tension members), using the same R, H, θ, and member-diameter parameterization. Rather than fabricating only the top-ranked geometries, we deliberately span the predicted performance range: a few designs the surrogate predicts to be *worst*-performing, a few *mediocre*, and a few *best*-performing are each built in both material systems. Comparing the measured ordering of the PLA/TPU specimens against their hollow-aluminum/stainless-steel counterparts tests whether both systems preserve the same rank ordering of energy-absorption performance, which would indicate that the optimization transfers across material systems rather than overfitting to the printed polymers. Table 4 reports the planned rank-ordering comparison and Fig. 7 shows the assembled metal-analog and printed-polymer specimens for each performance tier.

4 Results

4.1 Convergence of the BO Loop. Figure 8 illustrates the intended convergence comparison.

4.2 Pareto-Optimal Designs. Figure 9 shows the form the Pareto-front presentation will take.

4.3 Reproducibility Across Print Replicates.

5 Discussion

Topics to address:

- Comparison to the simulation-driven multifidelity framework of Mo et al. [17] and the single-condition fabrication study of Pajunen et al. [11].

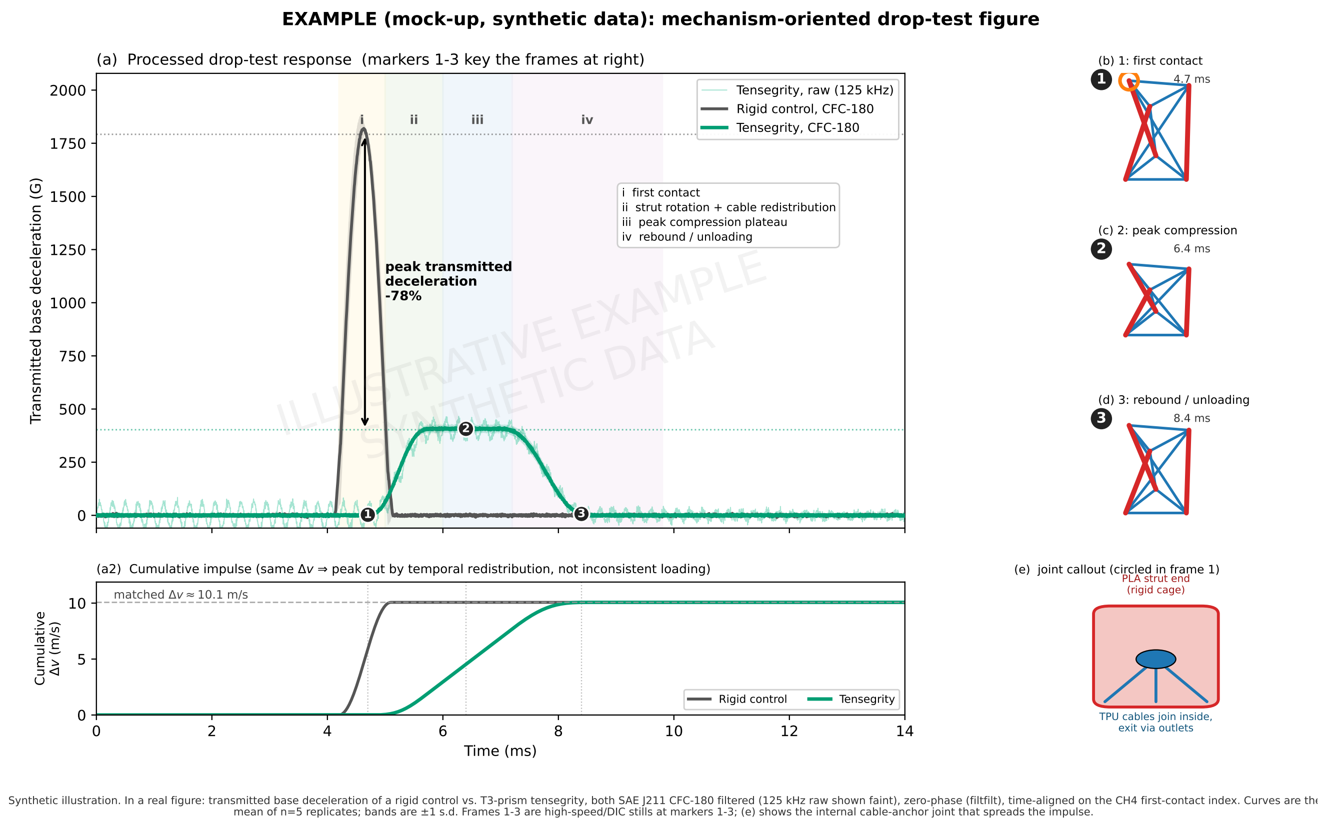


Fig. 4 ***Illustrative example (synthetic data).*** **Template for the mechanism-oriented drop-test figures planned for the Results: (a) processed deceleration response of a rigid control vs. a tensegrity specimen (SAE J211 CFC-180 filtered), with event-based phase shading and numbered markers keyed to the schematic frames (b)–(d); (a2) cumulative-impulse subpanel showing the two traces transfer the same mass-normalized Δv, so the lower tensegrity peak reflects temporal redistribution rather than inconsistent loading; (e) the internal-anchor joint load path. Curves are synthetic and impulse-consistent (control anchored to the documented ~1792 G CFC-180 peak; tensegrity shoulder scaled to the same impulse), pending real high-rate channels and camera frames. Generated by `scripts/figures/mechanistic_data_figure_example.py`.**

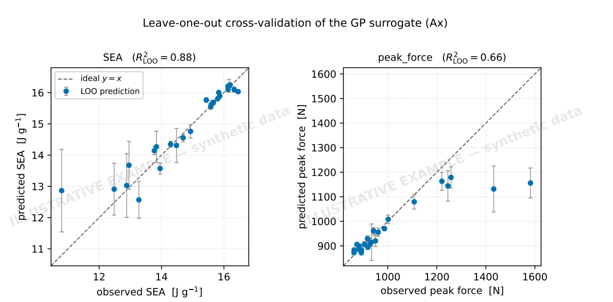


Fig. 5 ***Illustrative example (synthetic data).*** **Leave-one-out cross-validation of the Gaussian-process surrogate—predicted vs. observed specific energy absorption (SEA) and peak transmitted force, with model error bars and R^2_{LOO}—obtained from a real Ax/BoTorch run on synthetic objective values. To be regenerated from campaign data.**

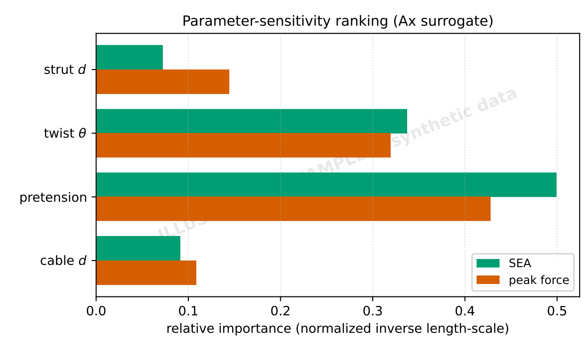


Fig. 6 ***Illustrative example (synthetic data).*** **Parameter-sensitivity ranking (model length-scale–based importances per objective; cf. SAASBO inverse length-scales or Sobol indices) of the design variables in Table 2, from the same Ax/BoTorch run as Fig. 5.**

- Practical considerations: TPU–PLA interface durability under cyclic loading; sensitivity of optimal designs to print parameters (extrusion temperature, layer height); transferability to TPU/PETG or other high-temperature thermoplastics.
- Path forward toward integrated tread+lattice structures, miniaturization to crutch-tip envelopes (from the Edison crutch-tip synthesis), and field testing. Long-term crutch use is associated with substantial upper-extremity overuse injury

and ground-reaction-force impulses [47–51]; existing spring-loaded [52,53] and polymer-damped [54] tip designs reduce loading rate but not peak transmitted force, leaving an open design opportunity that an architected tensegrity-inspired insert is well-positioned to address. Recent open-source 3D-printed crutch work [55] and expert evaluations of orthopaedic caps [56] suggest a plausible translation path;

Table 4 ***Placeholder (data pending).*** **Planned rank-ordering validation: representative T_3-prism designs predicted to be worst-, mid-, and best-performing, each fabricated as a printed PLA/TPU specimen and as a hollow-aluminum-rod / stainless-steel-cable metal analog. Entries record the measured performance metric (e.g. SEA or peak transmitted force) and the resulting rank within each material system; agreement of the two rankings indicates the optimization transfers across material systems.**

Predicted tier	Design ID	PLA/TPU (rank)	Al/SS (rank)
Worst	*D–w1*	–	–
Worst	*D–w2*	–	–
Mediocre	*D–m1*	–	–
Mediocre	*D–m2*	–	–
Best	*D–b1*	–	–
Best	*D–b2*	–	–

Figure placeholder: Assembled validation specimens for each predicted performance tier (worst / mediocre / best), shown as printed PLA/TPU builds alongside their hollow-aluminum-rod and stainless-steel-threaded-cable metal analogs, with callouts marking the aluminum compression members, the threaded steel tension cables, and the strut-end cable anchors. Photographs to be added once the specimens are fabricated.

Fig. 7 ***Placeholder.*** **Assembled validation specimens for each predicted performance tier (worst / mediocre / best), shown as printed PLA/TPU builds alongside their hollow-aluminum-rod and stainless-steel-threaded-cable metal analogs, with callouts marking the aluminum compression members, the threaded steel tension cables, and the strut-end cable anchors. Photographs to be added once the specimens are fabricated.**

adjacent footwear and orthotic lattice studies [57–59] provide additional transferable design heuristics.

5.0.1 *Limitations.*.

6 Conclusions

We have presented an experiment-driven Bayesian-optimization framework for designing multi-material 3D-printed tensegrity-inspired energy absorbers using PLA struts and TPU tension elements. The framework parameterizes a family of unit cells, fabricates each candidate on a single multi-material FDM platform with TPU tendons anchored inside the PLA strut ends, characterizes the response under quasi-static compression and drop-weight impact, and updates a Gaussian-process surrogate [21] that recommends the next batch of designs. The approach is broadly applicable to architected absorbers whose constitutive response is difficult to calibrate from first principles and where physical experiments—rather than simulations—are the natural ground truth.

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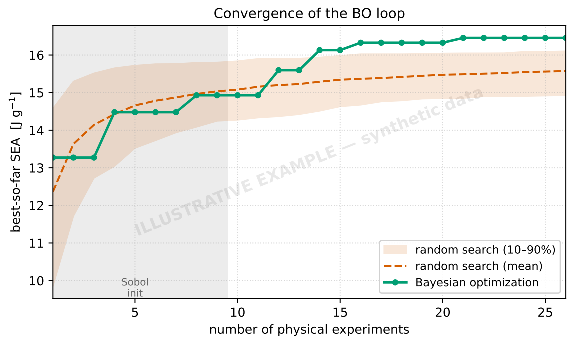

Fig. 8 ***Illustrative example (synthetic data).*** **Best-so-far SEA vs. number of physical experiments for the Ax/BoTorch loop against an independent random-search baseline; the shaded region is the $n = 9$ Sobol initialization. Generated on synthetic objective values; to be replaced with campaign results (with confidence intervals over random seeds, and Pareto hypervolume for the multi-objective runs).**

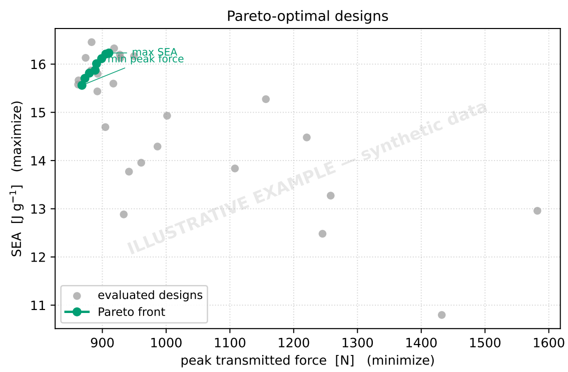

Fig. 9 ***Illustrative example (synthetic data).*** **Pareto front in (peak transmitted force, SEA) space with all evaluated designs, from the Ax/BoTorch loop on synthetic objective values; representative geometries to be annotated once campaign data are available.**

Conflict of Interest

The authors declare no conflict of interest.

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Table 5 ***Placeholder.*** **Mean and CV of SEA, peak force, and compaction efficiency across *n* replicate prints for the top-*k* designs.**

Column A	Column B	Column C
(table contents pending)		

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